MSET003

Question 1.
[Tg:@abusat_760] The dive tank managers at Seaside SCUBA Center try to keep their recreational SCUBA tanks filled with air at a pressure of approximately 3,000 pounds per square inch (psi). The maximum acceptable pressure for a tank is 3,300 psi, and the minimum acceptable pressure is 2,600 psi. Which inequality expresses the complete range of acceptable pressures, P, in psi, for the SCUBA tanks?

A) $|P - 2,950| \leq 350$

B) $|P - 2,950| \geq 350$

C) $|P - 3,000| \leq 300$

D) $|P - 3,000| \geq 400$

Question 2.
[Tg:@abusat_760] If $4(2^{y-2}) = 32^x$, what is the value of $\frac{y}{x}$?

Question 3.
[Tg:@abusat_760] Which of the following expressions cannot have a positive value for any real value of x?

A) $1 - |4 + x^2|$

B) $4 - |1 + x^2|$

C) $1 - |4 - x^2|$

D) $4 - |1 - x^2|$

Question 4.

$$-3(a-1)^2 + 2(a-1)$$

[Tg:@abusat_760] Which of the following is equivalent to the expression above?

A) $-a^2 + 2a - 1$

B) $-3a^2 + 2a$

C) $-3a^2 + 5a - 2$

D) $-3a^2 + 8a - 5$

Question 5.

[Tg:@abusat_760] If $x > 1$, which of the following is equivalent to $\dfrac{1 + \dfrac{1}{x+1}}{1 - \dfrac{1}{x-1}}$?

A) $\dfrac{(x-1)(x+2)}{(x+1)(x-2)}$

B) $\dfrac{(x-1)(x-2)}{(x+1)(x+2)}$

C) $\dfrac{(x+1)(x-2)}{(x-1)(x+2)}$

D) $\dfrac{(x+1)(x+2)}{(x-1)(x-2)}$

Question 6.
[Tg:@abusat_760] If $p(x)$ is a quadratic function and $p(-1) = p(7) = 2$, which of the following could be the coordinates of the vertex of the parabola formed when $y = p(x)$ is graphed in the xy-plane?

A) $(0, 8)$

B) $(4, 3)$

C) $(6, 7)$

D) $(3, 9)$

Question 7.

$$b = \frac{x+w}{y+w}$$

[Tg:@abusat_760] Given the formula above, which choice gives the formula for w in terms of x, y, and b?

A) $w = \dfrac{x-b}{y-b}$

B) $w = \dfrac{x-by}{b-1}$

C) $w = \dfrac{x-y}{b-x}$

D) $w = \dfrac{x-by}{b}$

Question 8.
[Tg:@abusat_760] If $a - 3b = 5$, what is the value of $\dfrac{3^a}{27^b}$?

A) 3^{-5}

B) 27^{-5}

C) 3^5

D) The value cannot be determined from the information given.

Question 9.
[Tg:@abusat_760] If $m = 2^k$, which of the following is equivalent to $(\dfrac{8}{m})^2$?

A) 2^{6-2k}

B) 2^{2k-6}

C) $2^{\frac{6}{k}}$

D) $2^{\frac{k}{6}}$

Question 10.
[**Tg:@abusat_760**] Which of the following is equal to $\sqrt[3]{x^7}$ for all values of x?

A) $x^{\frac{3}{7}}$

B) $x^{\frac{7}{3}}$

C) x^{10}

D) x^{21}

Question 11.

$$(ab^2c^2 + abc - a^2b^2c) - (a^2b^2c - ab^2c^2 + abc)$$

[**Tg:@abusat_760**] Which of the following is equivalent to the expression above?

A) $2abc - 2a^2b^2c$

B) $2abc + 2a^2b^2c$

C) $2ab^2c^2 - 2a^2b^2c$

D) $2ab^2c^2 + 2a^2b^2c$

Question 12.
[**Tg:@abusat_760**] If $\dfrac{1}{3}m + \dfrac{1}{12}p = 12$, what is the value of $4m + p$?

A) 36

B) 48

C) 72

D) 144

Question 13.

x	2	3	4	5	6
$f(x)$	5	4	3	2	1
$g(x)$	2	6	5	4	3

[**Tg:@abusat_760**] The table above shows some values of the functions $g(x)$ and $f(x)$ for particular values of x. If b is a number such that $g(f(b)) = 5$, which of the following could be a value of b?

A) 3

B) 4

C) 5

D) 6

Question 14.
[**Tg:@abusat_760**] Which of the following is equivalent to the quotient of the expressions $4a^2 - 1$ and $2a + 1$?

A) $a^2 + a$

B) $a - 1$

C) $2a - 1$

D) a^3

Question 15.
[**Tg:@abusat_760**] If $m^2 + n^2 = z$ and $mn = y$, which of the following is equivalent to $4z - 8y$?

A) $(2m - n)^2$

B) $(2m + 2n)^2$

C) $(2m - 2n)^2$

D) $(2m + 2n)^3$

Question 16.
[**Tg:@abusat_760**] Which of the following is equivalent to $25^{\frac{3}{4}}$?

A) $\sqrt[3]{25}$

B) $\sqrt[4]{25}$

C) $5\sqrt{5}$

D) $2\sqrt{5}$

Question 17.
[**Tg:@abusat_760**] Which of the following is equivalent to $\left(3a - \frac{b}{2}\right)^2$?

A) $9a^2 - \frac{b^2}{2}$

B) $9a^2 - \frac{b^2}{4}$

C) $9a^2 - 3ab + \frac{b^2}{4}$

D) $9a^2 - 2ab + \frac{b^2}{4}$

Question 18.
[Tg:@abusat_760] If $a^{\frac{b}{4}} = 81$ for positive integers a and b, what is one possible value of b?

Question 19.
[Tg:@abusat_760] Which of the following is equivalent to $(\sqrt{x} - \sqrt{y})^{\frac{2}{3}}$ where x and y are positive real numbers?

A) $(x - y)^3$

B) $\sqrt[3]{x - y}$

C) $(x - 2xy + y)^{\frac{1}{3}}$

D) $\sqrt[3]{x - 2\sqrt{xy} + y}$

Question 20.
[Tg:@abusat_760] If $a + b = 3$ and $a - b = -2$, what is the value of $a^2 - b^2 + 6$?

Question 21.

$$w = 2x^2$$
$$y = -3x^2$$

[Tg:@abusat_760] The equations above define w and y in terms of x. What is $2w - 3y$ in terms of x?

A) x^2

B) $-x^2$

C) $-13x^2$

D) $13x^2$

Question 22.
[Tg:@abusat_760] If m, t, and $c \geq 0$ and $m = \frac{\sqrt{1+c^2}}{t}$, what is c in terms of m and t?

A) $c = \sqrt{m^2t^2 - 1}$

B) $c = \sqrt{1 - m^2t^2}$

C) $c = \sqrt{1 + \frac{m^2}{t^2}}$

D) $c = \sqrt{\dfrac{m^2}{t^2} - 1}$

Question 23.
[Tg:@abusat_760] The total energy of an electron in orbit is the sum of its potential energy e^2r^{-2}, in joules, and its kinetic energy $-\dfrac{1}{2}e^2r^{-2}$, in joules. Which of the following gives the total energy?

A) $\dfrac{1}{2r^2}e^2$

B) 0

C) $-\dfrac{3}{2r^2}e^2$

D) $\dfrac{1-r}{2r^2}e^2$

Question 24.
[**Tg:@abusat_760**] Which of the following is equivalent to $k^{\frac{5}{6}}(k^{\frac{5}{6}})^{\frac{2}{5}}$?

A) $\sqrt{k^5}$

B) $\sqrt[6]{k^3}$

C) $\sqrt[6]{k^7}$

D) $\sqrt[6]{k^8}$

Question 25.
[**Tg:@abusat_760**] If $3(h-2)=3(t+2)$, what is h in terms of t?

A) $h=\dfrac{3t+3}{3}$

B) $h=t+4$

C) $h=\dfrac{3t+10}{3}$

D) $h=\dfrac{4t+11}{3}$

Question 26.
[**Tg:@abusat_760**] If $25x^2y^2=49$ and $xy>0$, what is the value of $15xy$?

A) 7

B) 20

C) 21

D) 35

Question 27.
[**Tg:@abusat_760**] If $x>0$ and $\sqrt{x\sqrt[3]{x}}=x^a$, what is the value of a?

A) $\dfrac{2}{3}$

B) $\dfrac{3}{4}$

C) $\dfrac{5}{2}$

D) $\frac{7}{4}$

Question 28.
[**Tg:@abusat_760**] The surface area, S, of a cylinder with a radius of 4 is defined by $S = 2\pi r^2 + 2\pi rh$, where h is the height of the cylinder. If the equation is rewritten in the form $h = \frac{S}{x\pi} - y$, where x and y are constants, what is the value of x?

Question 29.
[**Tg:@abusat_760**] If $\sqrt[3]{x} - \sqrt[3]{-8} = \sqrt[3]{-27}$, what is the value of x?

A) 125

B) 32

C) -32

D) -125

Question 30.
[**Tg:@abusat_760**] Which of the following expressions is equivalent to $\frac{(x-3)^2 - 9}{x^2 + 2x}$, where $x \neq 0, -2$?

A) $\frac{x+3}{x+2}$

B) $\frac{x-6}{x+2}$

C) $\frac{x-6}{x}$

D) $\frac{x+2}{x-6}$

Question 31.
[**Tg:@abusat_760**] For a car driving m miles on a highway, Toll A's cost, in dollars, can be modeled by $F(m) = 0.25m + 2.5$. Toll B's cost, in dollars, can be modeled by $H(m) = 0.30m + 3.5$. How much greater is $H(20)$ than $F(20)$?

A) 0

B) 0.5

C) 1.5

D) 2

Question 32.
[**Tg:@abusat_760**] If $\frac{3a}{5b} = \frac{12}{35}$, what is the value of $\frac{a}{b}$?

A) $\frac{7}{4}$

B) $\frac{4}{7}$

C) $\frac{12}{35}$

D) $\frac{60}{105}$

Question 33.
[**Tg:@abusat_760**] A serving of food X provides 52 calories and 14 grams of carbohydrates, while a serving of food Y provides 89 calories and 23 grams of carbohydrates. Which of the following combinations of X and Y servings contains more than 290 calories but less than 85 grams of carbohydrates?

A) 2 X servings and 2 Y servings.

B) 2 X servings and 3 Y servings.

C) 3 X servings and 2 Y servings.

D) 4 X servings and 1 Y serving.

Question 34.

$$2(10\sqrt{ax} - 5) = 2\sqrt{ax} + 26$$

[**Tg:@abusat_760**] Based on the equation above, what is the value of $2 - \sqrt{ax}$?

A) -2

B) -1

C) 0

D) 1

Question 35.
[**Tg:@abusat_760**] The amount of money, P, in a bank account earning simple interest can be determined using the formula $P = I(1 + rt)$, where I is the initial investment, r is the interest rate, and t is the time in years. Which formula shows the rate in terms of P, I, and t?

A) $r = \frac{P - I}{It}$

B) $r = \frac{P + I}{It}$

C) $r = \frac{P}{I(I + t)}$

D) $r = \frac{P - I}{I - t}$

Question 36.

$$\frac{5}{x-1}+\frac{8}{2(x-1)}$$

[**Tg:@abusat_760**] Which of the following expressions is equivalent to the one above, where $x \neq 1$?

A) $\dfrac{9}{x-1}$

B) $\dfrac{14}{x-1}$

C) $\dfrac{15}{2x-2}$

D) $\dfrac{21}{2x-2}$

Question 37.
[**Tg:@abusat_760**] For a positive real number x, where $x^8 = 2$, what is the value of x^{24}?

A) $\sqrt[3]{24}$

B) 4

C) 6

D) 8

Question 38.
[**Tg:@abusat_760**] Which of the following is an equivalent form of $\sqrt[3]{f^{6a}k^2}$, where $f > 0$ and $k > 0$?

A) $f^{\frac{1}{3a}}k^{-1}$

B) $f^{\frac{1}{2a}}k^{\frac{3}{2}}$

C) $f^{3a}k^{-1}$

D) $f^{2a}k^{\frac{2}{3}}$

Question 39.

$$|5-x| = 4$$

[**Tg:@abusat_760**] The value of the solution to the equation above is 1, what is the value of the other solution?

Question 40.
[**Tg:@abusat_760**] If $2\sqrt{2x} = a$, what is $2x$ in terms of a?

A) $\frac{a}{2}$

B) $\dfrac{a^2}{4}$

C) $\dfrac{a^2}{2}$

D) $4a^2$

Question 41.
[Tg:@abusat_760] When 9 is increased by $3x$, the result is greater than 36. What is the least possible integer value for x?

Question 42.
[Tg:@abusat_760] If $x \neq -1$, what is the value of $\left(\dfrac{1}{x+1}\right)(2+2x)$?

Question 43.

$$\frac{1}{2x+1}+5$$

[Tg:@abusat_760] Which of the following is equivalent to the expression above for $x > 0$?

A) $\dfrac{2x+5}{2x+1}$

B) $\dfrac{2x+6}{2x+1}$

C) $\dfrac{10x+5}{2x+1}$

D) $\dfrac{10x+6}{2x+1}$

Question 44.

$$(4x+4)(ax-1)-x^2+4$$

[Tg:@abusat_760] In the expression above, a is a constant. If the expression is equivalent to bx, where b is a constant, what is the value of b?

A) -5

B) -3

C) 0

D) 12

Question 45.
[Tg:@abusat_760] If the average of a and b is x, the average of b and c is $2x$, and the average of a and c is $3x$, what is the average of a, b, and c, in terms of x?

A) $\dfrac{2x}{3}$

B) $\dfrac{5x}{3}$

C) $2x$

D) $6x$

Question 46.
[Tg:@abusat_760] If $(x-a)(x-b)=x^2-9x+7$ for all values of x, what is the value of $a+b$?

Question 47.
[Tg:@abusat_760] If $f(x)=2x+1$, then what is $\dfrac{f(x+h)-f(x)}{h}$?

A) $\dfrac{1}{2}$

B) 2

C) $\dfrac{h-1}{h}$

D) $\dfrac{2h-2}{h}$

Question 48.
[Tg:@abusat_760] If $\dfrac{a^2+2ab+b^2}{a^2-b^2}=2(a+b)$, what is the value of $a-b$?

A) 1

B) $-\dfrac{1}{2}$

C) 2

D) $\dfrac{1}{2}$

Question 49.
[Tg:@abusat_760] If $f(n-1)=13+4n$ for all values of n, what is the value of $f(3)$?

Question 50.
[Tg:@abusat_760] If the rational expression $\dfrac{9x^2-4}{x+1}$ is rewritten in the equivalent form $\frac{5}{x+1}+A$, what is A in terms of x?

A) $x-1$

B) $x+1$

C) $9x-9$

D) $12x-3$

Answer Sheet

1	2	3	4	5	6	7	8	9	10
A	5	A	D	A	D	B	C	A	B

11	12	13	14	15	16	17	18	19	20
C	D	A	C	C	C	C	4,8,16	D	0

21	22	23	24	25	26	27	28	29	30
D	A	A	C	B	C	A	8	D	B

31	32	33	34	35	36	37	38	39	40
D	B	D	C	A	A	D	D	9	B

41	42	43	44	45	46	47	48	49	50
10	2	D	B	C	9	B	D	29	C